

MELAKU ALEHEGN

Full Stack AI / Machine Learning Engineer

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Summary

Full Stack AI Software Engineer with experience designing and deploying production-grade machine learning and LLM-based systems. Strong focus on end-to-end pipelines, data-intensive feature engineering, and integrating AI models into real-world products. Comfortable working across data, backend, and infrastructure layers to deliver reliable, scalable solutions.

Experience

Wicked Analytics, Machine Learning Engineer 02/2026 – Present | Remote

- Designed, trained, and deployed a 16-week ML demand forecasting model trained on multi-year historical Walmart sales and store-level data for retail-supplier clients, producing per-item weekly predictions that drive replenishment decisions.
- Built Airflow-orchestrated pipelines that pull daily sales, demand forecasts, supply plans, and supplier-side production data into Snowflake, transform them through DBT models, and distribute weekly outputs to client-facing storage and BI dashboards.
- Modeled the core data layer in DBT and Snowflake: per-client transformation logic for store-level sales, demand forecasts, supply plans, and supplier inventory.
- Built a full-stack demand planning web application (React, FastAPI, PostgreSQL) for planners: interactive item list, editable per-week order plans, projected-inventory cascade computed on the fly, and multi-vendor support, replacing legacy spreadsheet-only workflows.
- Integrated automated weekly Excel deliverables (forecasts, supply plans, valid-store reports) into existing client workflows with stable naming and consistent file locations, so planners can drop the outputs straight into established planning workbooks without disrupting their cadence.

Arifpay, Data Scientist 01/2025 – 05/2026 | Addis Ababa, Ethiopia

- Built and deployed machine learning models for payment fraud detection, working with high-volume transactional data and severe class imbalance
- Designed behavioral and temporal feature pipelines to improve fraud signal quality, contributing to a 15% reduction in fraud incidents
- Developed a merchant credit scoring system using FastAPI, improving lending decision accuracy and increasing approval confidence by around 40%
- Implemented continuous learning pipelines using MLflow and Celery to retrain and monitor models in production
- Built and maintained Airflow DAGs for ML workflows, enabling reliable scheduling and monitoring and helping prevent an estimated \$100K in potential annual losses
- Designed Metabase dashboards for executive-level KPIs (revenue, churn, acquisition, efficiency), enabling data-driven decision-making across the business
- Built Retrieval-Augmented Generation (RAG) systems over internal documents and operational data to support internal analytics and decision support workflows, focusing on reliability and real-world usability

Openstack, Software Developer Intern 12/2023 – 04/2024 | Remote

- Enhanced the UI workflow of the OpenStack Manila project using Django views, streamlining the share network creation process and improving usability.

- Increased test reliability by achieving 90% unit test coverage and introducing pre-commit and tox configurations to standardize development workflows.
- Collaborated with OpenStack core reviewers to develop, refine, and merge patches within release cycles in a large-scale open-source environment.

Nedamco Africa, *Cloud Consultant*

12/2023 – 03/2024 | Addis Ababa, Ethiopia

- Consulted clients across agritech, hospitality, and pharmaceutical industries on cloud adoption strategies using AWS and Azure platforms.
- Assisted in designing scalable cloud architectures and guided clients on selecting services for compute, storage, networking, monitoring, and cost optimization.
- Supported cloud migration and deployment planning while working with AWS services including EC2, S3, IAM, RDS, and CloudWatch.
- Participated in stakeholder discussions and prepared technical recommendations, architecture documentation, and implementation guidance aligned with business requirements.

MMCY Tech, *Full Stack Developer Intern*

07/2023 – 01/2024 | Addis Ababa, Ethiopia

- Developed a full-stack Applicant Tracking System (ATS) web app using Vue.js, Express.js, and Tailwind CSS, designing schemas and API endpoints for user authentication and job management
- Implemented core ATS features including login, candidate listings, job listings, and an admin dashboard, enhancing recruiter productivity with streamlined workflows
- Collaborated in an agile environment to build and maintain software applications and databases, ensuring timely delivery of project milestones

Projects

Verba, *Full Stack Multi-tenant RAG platform*

Stack: FastAPI, Qdrant, Celery, React, Gemini API

Async ingestion (parse→chunk→embed), retrieval with MMR reranking + hybrid (keyword+vector) search, conversational query rewriting, token-streamed answers with source citations, and a strict grounding threshold that refuses to hallucinate. Hardened with OAuth, per-IP rate limiting, structured logging, readiness probes, and a 140+ test suite with CI.

Chimera, *Autonomous multi-agent system*

Stack: Java 21, Gemini API, PostgreSQL, MCP

Manager/Worker/Judge agent architecture with LLM-backed reasoning, event-sourced memory, a custom Model Context Protocol client written from scratch, and autonomous publishing to social media.

Education

Addis Ababa Science & Technology University

09/2016 – 09/2021 | Addis Ababa, Ethiopia

BSc in Electrical and Computer Engineering

Skills

Applied AI / ML (Python, Pandas, NumPy, Scikit-learn, TensorFlow) | **Data & MLOps** (Apache Airflow, MLflow, Celery, Kafka, Snowflake, Data Pipelines) | **Backend & Deployment** (FastAPI, Java, Flask, Docker, REST APIs, Redis, Node.js, CI/CD) | **LLMs & Search** (LangChain, Vector Databases (ChromaDB, Qdrant, PgVector), MCP, Multi-Agent Systems) | **Cloud & Analytics** (AWS, GCP, Metabase, PowerBI, Azure)